Classify Objects with VLA Grasp Network

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Abstract: In recent years, vision-language-action models (VLAs) have become dominant in robotics domain. In this work, we leverage both visual and textual information to build a versatile grasp network. To test our algorithm, we set up an experimental environment where objects of identical color were grouped and placed on the same plate. Evaluated in our environment, our framework demonstrates superior grasping performance.

Keywords: VLA, Imitation learning, Grasping

1. INTRODUCTION

Driven by the remarkable advances in vision language models (VLMs) [1, 3–5], which integrate visual and textual information have recently delivered impressive performance across a wide range of domains. Equipped with vast repositories of general knowledge, these models have given rise to a host of approaches—also extending robotic task planning, navigation, and manipulation. Within these approaches, VLMs decompose high-level vision and language instructions into structured sequences of primitive skills, enabling robots to handle complex tasks effectively.

Meanwhile, advances in vision language action models (VLAs) have ushered in a new paradigm in robotics and embodied AI. Although VLMs have demonstrated strong performance in these domains, they primarily function in nonphysical settings, which makes them difficult to apply directly to robots interacting with the physical world. But VLAs process robotic visual observations and human instructions to directly generate robot actions. Building upon these foundations, we further exploit VLA model for long-horizon manipulation tasks, where semantic object classification precedes and guides sequential grasp planning. Our approach enables robots to execute vision-text-guided manipulation strategies in complex and cluttered environments.

In this paper, we propose our algorithm that adopts a diffusion model to generate actions conditioned on visual and textual information, leveraging a FiLM-based feature modulation [6]. We leverage the pre-trained VLM PaliGemma2 [1] to extract multi-modal embeddings, which we then use to condition our action generation model. This allows our model to intuitively select the right objects and achieve higher performance in complex environments. Our contributions are:

- We propose a novel VLA architecture that integrates pre-trained VLM with a diffusion-based action generation model, with FiLM for efficient multi-modal conditioning.
- In our experiments, we first classified four objects by color and executed a long-horizon grasping task by generating instructions for each required subtask. Our approach achieved a high success rate.

2. RELATED WORK

Vision-Language-Action Models. Following the success of pre-trained VLMs, several works explore leveraging additional action expert to generate multi-modal actions with high fidelity. Our work is most closely related to recently proposed VLA models, which use pre-trained VLMs that fine-tuned for robot control. RT-2 [9] and OpenVLA [10] propose to leverage pre-trained VLM to exploit the rich knowledge from Internet dataset. However, limited by the scale of real-world robotic data, existing VLA models mainly rely on in-domain post-training for deployment. Concurrent work, $\pi_{0.5}$ [11], proposes improving generalization by leveraging multi-modal web data and cross-embodiment data, enabling direct out-of-the-box deployment. Our work uses a custom dataset to further train the action generation model and then performs only the tasks aligned with its capabilities.

Imitation Learning for Robotic Manipulation. Imitation learning allows a robot to learn directly from experts. Behavioral cloning (BC) is one of the simplest imitation learning algorithms, casting imitation as supervised learning from observations to actions. Many works have then sought to improve BC, by incorporating history with various architectures, using a different training objective, and including regularization.

Building on these enhancements, diffusion-based approaches have recently been proposed to better capture complex action distributions. Diffusion Policy [7] casts policy learning as a denoising diffusion process, iteratively refining noisy action samples into coherent control commands that more faithfully reproduce expert behaviors.

In parallel, transformer-based architectures have been introduced to address long-horizon dependencies in imitation learning. The Action Chunking Transformer (ACT) [8] segments demonstration trajectories into fixed-length "chunks" of consecutive actions and uses a transformer backbone to model temporal correlations both within and across chunks, enabling scalable planning and robust generalization over extended task sequences.

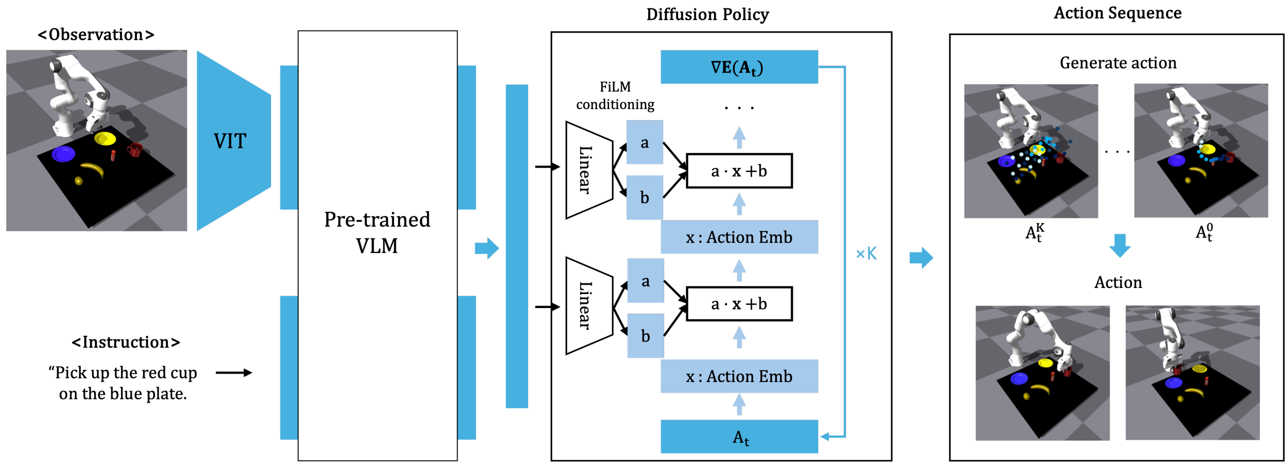

Fig. 1.: **Overview of our framework.** We use the observation image and instruction text are fed into pre-trained VLM model. and its output features condition the policy generation module.

3. METHOD

3.1 Backbone pre-trained model

We leverage PaliGemma2 as the backbone for our pre-trained VLM. It combine with the SigLIP [2] vision encoder and 2B Gemma [13] language model. It matches the performance of much larger prior VLMs consisting of a range of different vision encoders and language models.

3.2 Feature-wise Linear Modulation (FiLM)

Feature-wise linear modulation (FiLM) model is a effective feature conditioning model that injects external information directly into the different network. First, in FiLM network, the vector that inherent the context of feature passes through a mlp layer to produce two sets of parameters, commonly called γ (scale) and β (shift). These parameters are then applied channel-wise to the target feature maps via the simple operation:

$$FiLM(\mathbf{x_i}|\gamma_\mathbf{i}, \beta_\mathbf{i}) = \gamma_\mathbf{i}\mathbf{x_i} + \beta_\mathbf{i}$$

Here, subscript refers to the i^{th} input, and x is the neural network activation. By inserting FiLM layers, the network can dynamically transfer its context to a different model. This approach is highly parameter-efficient, since each channel in each layer can be adjusted independently.

3.3 Diffusion Policy

We adopt the CNN-based Diffusion Policy in our model, which consists of the 1D temporal CNN [12]. This model employs the denoising diffusion process over the action sequences. At each time step t, the model executes K denoising iterations to generate the corresponding action sequence. Then it executes only a subset of these actions.

4. EXPERIMENTS

4.1 Environments

We present a series of experiments in which we classify four different objects by placing objects of the same color onto plates of the corresponding color. We will use the four colors red, blue, yellow, and green for both the objects and their corresponding plates. Note that when setting up the environment, we grouped the four objects into two color categories, pairing them two by two. We used three objects of each color (see Table 1 for the full list), and the plate colors were randomly assigned, irrespective of the object colors. We conducted the classify environments in IsaacGym [14]. The task is complete once the robot places all objects onto the two plates corresponding to their colors.

Table 1. Objects in each color category.

Color	Objects
Red	Apple, Sugar bottle, Mug cup
Blue	Cleanser, Cup, Mug cup
Yellow	Lemon, Banana, Cup
Green	Pear, Mug cup, Cup

4.2 Results

Table 2. Objects in each color category.

Color	Success Rate
Red	93%
Blue	87%
Yellow	83%
Green	95%

Given the numerous combinations of colors and objects in our experiments, we evaluated the success rates for each color separately. For example, we calculate the success rate for yellow objects by averaging outcomes across pairings of yellow objects with red, blue, and green objects. Table 2 shows the success rates for the four colors, indicating that the green category achieved the highest success rate and the yellow category achieved the lowest. The VLM model generated appropriate instructions regardless of the color of the object. The difference in success rate seems attributable to object shape rather than color: the yellow category includes items that require varied orientations for grasping, whereas the green category consists of objects that can be grasped reliably in a single orientation.

5. CONCLUSION

This paper presented an effective VLA model that leverages pre-trained VLM model alongside a diffusion-based grasp action generation framework. Our system identifies target objects efficiently and generates optimal trajectories to perform precise grasping actions. Despite these advances, our approach has limitations about cost-efficient action generation and mechanisms for conditioning on visual and textual inputs. Future work will address these limitations by optimizing and enriching spatial understanding and semantic alignment.

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